

Novel Cognitive Functions Arise at the Convergence of Macroscale Gradients

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Abstract

■ Functions in higher-order brain regions are the source of extensive debate. Past trends have been to describe the brain in terms of a set of functional modules, especially posterior cortical areas, but a new emerging paradigm focuses on interactions between neighboring representations. In this review, we synthesize emerging evidence that a variety of novel functions in the higher-order brain regions are due to convergence. Convergence of macroscale gradients brings feature-rich representations into close proximity, presenting an opportunity for novel functions to arise. Using the TPJ as an example, we demonstrate that convergent areas have three properties, they: (1) are at the peak of the

processing hierarchy, (2) combine the most abstracted representations, and (3) are equidistant from other convergent areas. As information moves from primary sensory cortices to higher-order brain regions, it becomes abstracted and hierarchical. Eventually, these processing gradients converge at a point equally and maximally distant from their sensory origins. This convergence, which produces multifaceted cognitive functions, such as mentalizing another person's thoughts or projecting into a future space, parallels evolutionary and developmental characteristics of such regions, resulting in new cognitive and affective faculties. ■

INTRODUCTION

Higher-order cognitive functions are important for humans to navigate a complex world. Our daily interactions require contributions from these higher-order functions, for example, attention, memory, value computation, cognitive control, language, and social processes; all are computationally expensive processes that we carry out routinely but are only beginning to be understood. What we do know is that these functions are not allocated randomly in the brain. Some of these regions—including the TPJ, the medial pFC (mPFC), and the posterior cingulate cortex (PCC)—serve as hubs, commonly active in many tasks. Given this pattern, how do higher-order cognitive functions occur? Although a number of theories exist on how these functions arise (Man, Kaplan, Damasio, & Damasio, 2013), the mechanisms by which higher-order cognitive functions (e.g., theory of mind) occur are yet to be explored. We argue that higher-order cognitive functions arise through convergence.

One fundamental, but overlooked, aspect of understanding higher-order cognition is the role of lower-order sensory processing streams or gradients. Sensory processing areas are, by definition, devoted to processing a single sensory modality. These primary areas then provide inputs to produce the next, relatively more complex and abstracted representation in a neighboring cortex. Visual inputs, for example, are processed in the occipital lobe, beginning primarily with neurons that detect line

orientations in very small areas of the visual field (Hubel & Wiesel, 1962). Moving farther away from the primary visual cortex, receptive fields grow larger, lines are combined into longer arcs, and preferred stimuli become simultaneously more complex and relatively domain-general (Figure 1A). This progression along the visual processing hierarchy can also be described as a process of abstraction, stripping away finer details in favor of a general, more conceptual set of responses. A similar progression is present for auditory (Figure 1B) and somatosensory (Figure 1C) cortices. In aggregate, areas of the brain reaching from sensory areas to higher-order hubs can be viewed as hierarchical processing pipelines, such as visual systems, famously modeled in the work of Van Essen and Maunsell (1983) and well modeled in deep learning neural networks of vision (Kuzovkin et al., 2018; Wenliang & Seitz, 2018; Güçlü & van Gerven, 2015). Such progressions are anchored in primary sensory cortices and culminate in polymodal and highly abstracted representations. This description of overall brain structure mirrors an emergence of diffusion measures of information processing in the brain (Larivière et al., 2020; Paquola et al., 2019; Vázquez-Rodríguez et al., 2019). Where processing streams converge, these areas combine the converging information, and this mixture of information gives rise to multifaceted novel functions, characteristic of higher-order cognition.

Using the TPJ as an example, we present a framework of convergence of macroscale gradients. We argue that convergence provides a rich setting for information to be integrated in a manner that gives rise to multifaceted cognitive

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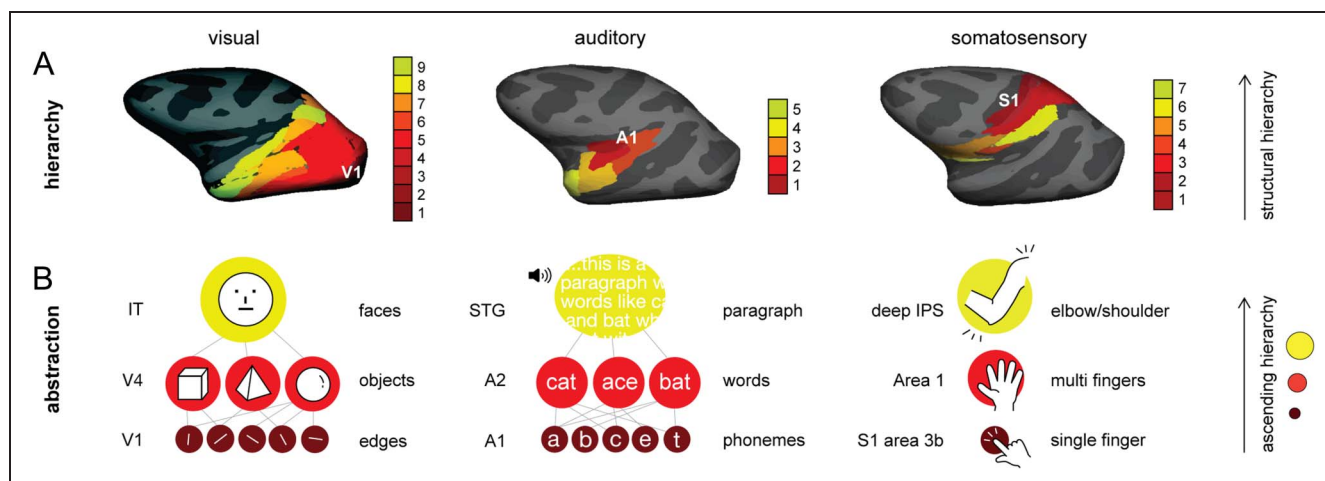


Figure 1. Cortical hierarchies parallel gradients of abstraction in the brain. (A) Cortical hierarchies for processing visual, auditory and somatosensory modalities are hierarchically organized, beginning with primary sensory (“red”) to higher-order regions (“yellow”; Wagstyl et al., 2015). Their hierarchical numeric scales are based upon prior literature on structural hierarchy: visual (Felleman & Van Essen, 1991), auditory (Barbas, 1986), and somatosensory (Felleman & Van Essen, 1991). (B) We acknowledge the discrepancy between a macaque brain and higher order abstract cognition that is performed in a human brain. The macaque brain is for visual purposes of the hierarchical organization of the brain, which is both functionally and structurally well-established. We do not claim that there is a one-to-one mapping of abstract content in the macaque brain; rather, we use the visualization from (A) to establish the hierarchical organization in a brain. From this, we expand on the idea that hierarchy enables abstraction, which is a unique feature of the human brain. Moving from primary sensory to higher-order areas, representations become more abstracted, as seen in this schematic of levels of abstraction. Primary sensory regions are associated with low-level features (i.e., edges, phonemes, and single finger joints) and smaller receptive fields. In contrast, higher-order regions are associated with abstract concepts (i.e., faces, paragraphs, and elbow/shoulder joints) and larger receptive fields. The abstracted levels in our schematic are based on prior literature: visual (Dicarlo, Zoccolan, & Rust, 2012), auditory (Kikuchi et al., 2010; Wessinger et al., 2001; Merzenich & Brugge, 1973), and somatosensory (Iwamura et al., 1993).

functions. We then focus on answering two questions: How does convergence arise in the brain, and what happens when processing streams converge? To answer these questions, we first overview the literature on TPJ function, focusing particularly on cognitive functions traditionally identified as single modular functions, such as attention, memory, and social processes. We discuss what happens in regions where these cognitive processes overlap by introducing studies that demonstrate that novel functions arise from a mixture of these single cognitive processes. We ask how this overlap occurs and describe characteristics of the brain that result in the convergence of macroscale gradients. After establishing the neural architecture that provides a setting for novel functions to arise, we lay out characteristics of these functions and review studies that demonstrate their emergence within the TPJ. We argue that representations in convergence zones are not merely an addition of contributing cognitive processes from lower in the hierarchy. Instead, these representations are novel in that they are combinations of the surrounding processing streams, modeling aspects of the world that cannot be constructed without all of the convergent information (e.g., predicting future responses of social agents). Coming from a convergence perspective, we describe functions within convergence zones as relatively “complex,” where new representations are formed by combinatoric mixtures of surrounding processes, in contrast to relatively simple representations that occur outside of convergence zones.

The principal reason for the complexity of functions within convergence zones extends from their location and topology. Situated at the apex of processing streams, the convenient location provides an opportunity to combine the most abstracted representations from a number of incoming processing streams. To illustrate the nature of complexity in convergence zones, recent work from Behrens et al. (2018) has shown that grid cell-like behaviors, indicators of representations in highly abstracted concept space, can be found within convergence zones (i.e., TPJ, ventral mPFC [vmPFC], and PCC). We illustrate that novel, complex representations of convergence zones result from three characteristics of macroscale gradients: (1) hierarchical organization, (2) abstraction, and (3) maximally and equidistant from primary sensory cortices, producing multifaceted combinations of cognitive functions. Toward the end of the section, we discuss how this convergence parallels evolutionary and developmental characteristics in such regions, resulting in new cognitive and affective faculties.

Getting Insight from the TPJ

There are several regions of the brain that support higher-order cognition. Various called convergence–divergence (Damasio, 1989), associative (Buckner & Krienen, 2013), amodal (Fairhall & Caramazza, 2013; Tamber-Rosenau, Dux, Tombu, Asplund, & Marois, 2013), transmodal (Mesulam, 1998), and polymodal (Baumgaertner,

Buccino, Lange, McNamara, & Binkofski, 2007), these brain regions have all been shown to participate in a wide range of cognitive processes, but with little agreement about the specific role each region plays. The TPJ is such a region (Bukowski, Lamm, Zeigler-Hill, & Shackelford, 2017). We review the debate over function in the TPJ, starting with single modular views, then describing a domain-specific view, in which single functions split the TPJ, and finally moving toward a more domain-general view.

It is important to note that individual studies have varying ways to define the TPJ. It is variably referred to as angular gyrus, intraparietal sulcus, STS, anterior TPJ, and posterior TPJ. This heterogeneity, not only in defining the region itself but also in assigning names to these regions, makes it difficult to pinpoint what we actually mean by the TPJ. To adopt a relatively unbiased and inclusive definition of the TPJ, here we use the neurosynth meta-analysis map of the TPJ.

Early work characterizing function in the TPJ was a series of contributions from different areas of cognitive neuroscience, each claiming that the TPJ was involved only in attention, memory, or social processes. Evidence that the TPJ was an attention module was based on neuroimaging during reorienting tasks—such as the Posner cueing paradigm (Mitchell, 2008; Konrad et al., 2005; Corbetta & Shulman, 2002) or oddball detection tasks (Fichtenholtz et al., 2004; Stevens, Skudlarski, Gatenby, & Gore, 2000; Downar, Crawley, Mikulis, & Davis, 2001)—demonstrating greater activation in the TPJ when attentional reorienting or attentional detection was required. Neuropsychologists bolstered the argument, noting that spatial neglect was commonly associated with TPJ damage. One example is the claim that damage to the right TPJ is associated with attentional deficits in the contralateral visual field (Friedrich, Egly, Rafal, & Beck, 1998). However, this was later shown to be a mislocalization, with the crucial region being slightly more anterior (Smith, Clithero, Rorden, & Karnath, 2013).

Memory—specifically episodic memory retrieval—was also thought to be a primary cognitive function associated with the TPJ. Tasks that involved old/new recognition recruited the TPJ, primarily for successfully retrieved old and successfully rejected new words (Konishi, Wheeler, Donaldson, & Buckner, 2000). Confidence in episodic retrieval also correlated with TPJ activation, where greater activation was observed for responses with higher confidence (Cabeza, Ciaramelli, Olson, & Moscovitch, 2008; Kim & Cabeza, 2007; Wagner, Shannon, Kahn, & Buckner, 2005).

Others have argued that the TPJ was a purely social region, based on neuroimaging results collected during theory of mind tasks, that is, inferring the mental state of other social agents (Kestemont, Vandekerckhove, Ma, Van Hoeck, & Van Overwalle, 2013; Saxe & Powell, 2006) in a variety of formats: theory of mind stories (Döhnell et al., 2012; Aichhorn et al., 2009; Kobayashi, Glover, &

Temple, 2007; Young, Cushman, Hauser, & Saxe, 2007; Saxe & Powell, 2006; Saxe & Kanwisher, 2003), inferring mental states from pictures (Rabin, Gilboa, Stuss, Mar, & Rosenbaum, 2010; Samson, Zysset, & Huber, 2008), eye gaze interpretation (Mitchell, Banaji, & Macrae, 2005; Platek, Keenan, Gallup Jr, & Mohamed, 2004; Baron-Cohen et al., 1999), and playing games with a social opponent (Powell, Grossi, Corcoran, Gobet, & García-Fiñana, 2017; Kircher et al., 2009; Gallagher, Jack, Roepstorff, & Frith, 2002). Distinct patterns in the right TPJ were also associated with autism, a clinical condition linked with social communication deficits, suggesting that the TPJ is a region involved in social processing (Lombardo, Chakrabarti, Bullmore, MRC AIMS Consortium, & Baron-Cohen, 2011). The involvement of TPJ in social processing was also demonstrated by TMS studies, where temporally disrupting the right TPJ affects reasoning about others' mental states (Young, Dodell-Feder, & Saxe, 2010).

These studies have only served to fuel debate over function in the TPJ. With all of the conflicting information, researchers began to question the single-module view of the TPJ, which posited that the TPJ is entirely devoted to one cognitive process. In fact, studies that included direct contrasts of attention and social tasks found that parts of the TPJ became active during both tasks, but that there were also subregions of the TPJ that seemed to be more responsive to one function than the other (Mitchell, 2008).

Hence, following the debate over a singular function for the TPJ, researchers proposed that the TPJ is fractionated, that is, that the region is a combination of single-function subunits. There are three lines of evidence supporting a fractionation view. First, within-study comparisons have found stronger activation in subregions of the TPJ for specific cognitive tasks (Hutchinson et al., 2014; Scholz, Triantafyllou, Whitfield-Gabrieli, Brown, & Saxe, 2009). For example, Scholz et al. (2009) found neighboring but distinct peak activation (6–10 mm spatial difference) for theory of mind and attentional reorienting tasks within participants. Second, via network parcellation, anatomical and functional heterogeneity has been identified within the TPJ. In one study, three subregions were identified using diffusion-weighted imaging, and from that, distinct functional connectivity was identified, lending support to the idea of disparate functions from disparate subregions (Mars et al., 2012). Strong correlation between diffusion-weighted imaging tracts and functional connectivity was found in TPJ subregions, suggesting a correspondence between anatomical and functional connectivity patterns of TPJ architecture (Wang et al., 2016). Lastly, task-based fMRI studies have found that different cognitive processes are localized in different parts of the TPJ. For example, three spatially distinct regions were identified for three different tasks: theory of mind, reorienting, and oddball detection (Kubit & Jack, 2013). Similarly, an independent component analysis with five different cognitive tasks—memory, social attention, reorienting, oddball detection, and theory of mind tasks—found five distinct activation

profiles (Igelström, Webb, Kelly, & Graziano, 2016), which support the view of TPJ being composed of multiple subregions.

In contrast to the fractionation view, others have argued for an “overarching view” of TPJ function (Cabeza et al., 2011). Such views hold that the TPJ is associated with a single function that serves as an ingredient for many tasks, with all contributing areas serving to fulfill that one function. For example, Geng and Vossel (2013) proposed that the TPJ enables “contextual updating,” which is a combination of postperceptual processes and top-down modulation. Others have proposed an overarching function of attention (Lückmann, Jacobs, & Sack, 2014) or an overarching cognitive mechanism in attentional control and mentalization (Schuwerk, Schurz, Müller, Rupprecht, & Sommer, 2017). This coordinated and holistic view of TPJ function is supported by a meta-analysis by Schurz, Radua, Aichhorn, Richlan, and Perner (2014), which finds areas of high consistency across a wide variety of social cognitive tasks. However, neither the fractionation nor the overarching view addresses the fact that the TPJ supports not only one of the proposed functions but instead an intersection of the proposed cognitive functions (Carter, Bowling, Reeck, & Huettel, 2012).

In fact, the nexus model of TPJ function proposes that novel cognitive processes emerge as single cognitive processes overlap (Carter & Huettel, 2013). This overlap produces a context for social predictions based on past experiences and eventually leads to the emergence of novel functions that would not be possible without input from neighboring areas. In other words, in contrast to the fractionation view, the nexus model posits that the TPJ is not a set of independent subregions but carries out novel and multifaceted functions that reflect characteristics of the neighboring processes. Moreover, in contrast to the overarching view, the nexus model hypothesizes that the

central portion of the TPJ does not support any single cognitive domain but instead combines information from each of these domains into a novel function.

Single Cognitive Domains Intersect the TPJ

There is, in fact, accumulating evidence that novel combinatorial functions are present in higher-order cognitive regions like the TPJ. Interestingly, support for convergence of functions comes from past studies originally meant to argue that the TPJ was a series of fractionated submodules. However, a different question could be asked: Is there evidence of overlap among different cognitive functions? There is indeed clear evidence of overlap when studies combine paradigms from different areas of cognitive neuroscience (Figure 2): memory and perception (Cabeza et al., 2011); theory of mind and attention (Krall et al., 2015; Mitchell, 2008); overlap of reorienting, empathy, and theory of mind (Decety & Lamm, 2007); and default mode and social processes (Padmanabhan, Lynch, Schaer, & Menon, 2017). Here, we review four articles characterizing cognitive processes of overlapping activation in cortical regions surrounding the TPJ.

Starting with one cognitive function, attention, Vossel, Geng, and Fink (2014) provide evidence of overlap between the two distinct networks (dorsal and ventral) within the TPJ. Early on, the TPJ was hypothesized as primarily serving as a bottom-up circuit breaker for ongoing actions. However, recent studies have shown that responses in the TPJ are temporally delayed when compared with the dorsal attention network regions, such as the FEF and intraparietal sulcus, making it difficult to imagine that the TPJ is solely an early bottom-up reorienting area. In fact, the TPJ combines information from separate systems into unique representations in the brain, supporting the argument for interaction; for example, the

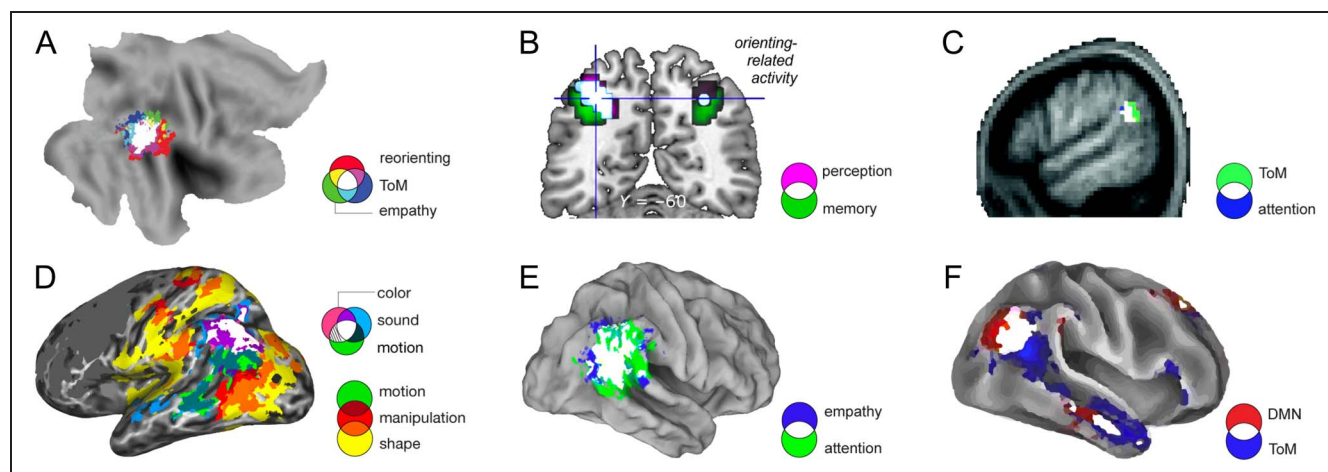


Figure 2. As a convergence zone, the TPJ shows overlapping activation of higher-order cognitive functions. Each color corresponds to the contrast mask within each study (A) reorienting, ToM, and empathy (Decety & Lamm, 2007). (B) perception and memory (Cabeza et al., 2011). (C) ToM and attention (Mitchell, 2008). (D) color, sound, motion, manipulation, shape of words (Fernandino et al., 2016). (E) empathy and attention (Decety & Lamm, 2007). (F) DMN and ToM (Padmanabhan et al., 2017). ToM, theory of mind; DMN, default mode network. (Overlap of the original figures has been recolored to white for visualization purposes.)

TPJ integrates contextual information for nontarget stimuli, construed as attentional control signals of the TPJ, formed only after interaction takes place with the dorsal and ventral attentional regions (DiQuattro & Geng, 2011). Geng and Vossel (2013) conclude that ventral and dorsal attention networks develop separately, but by the time these networks converge in the TPJ, there is flexible interaction between the two systems that is dependent on the nature of the task. This suggests a role for convergence zones that is more than a simple extension of any one of the converging processing streams. It suggests convergence zones mediate interactions between disparate functions.

Evidence of overlap among different cognitive functions also comes from research on attention and memory, which are traditionally thought to be independent. However, interactions between the two processes result in combined activation localized in convergence zones. The attention-to-memory model (Cabeza et al., 2008, 2011; Ciaramelli, Grady, & Moscovitch, 2008) posits that the dual attention network is also paralleled in memory processes, a circumstance in which the convergence framework predicts would produce new combinations of these functions. Cabeza et al. (2011) show that the superior parietal lobule performs strategic memory retrieval, a combination of memory and top-down attention from the dorsal streams. The inferior parietal lobule is activated by a combination of memory and bottom-up attention where those streams converge. From these results, it could be extrapolated that when these large-scale gradients of attention, memory, and perception meet, they produce unique processes in predictable locations.

In addition, there is evidence of statistical interaction between social and attention processes. Schuwerk et al. (2017) designed an attention-orienting task (Posner cueing task) in a social versus nonsocial context to isolate the region (or regions) of the brain that support a task, which requires interaction between processing streams. Univariate analysis revealed a significant interaction effect of attention and social conditions in the TPJ. The TPJ was differentially activated for a unique combination of attentional and social processes, which demonstrates that the overlap of two processes leads to superadditive, novel functions within the TPJ.

Thus far, by calling on evidence from studies with more than two traditional cognitive domains, we have demonstrated that overlap of cognitive functions in nexus regions creates a setting for new functions to arise. So, how does this overlap occur? How do cognitive functions converge within regions like the TPJ? In the next section, we describe the neural architectural characteristics that enable the convergence of what is called macroscale gradients.

How Does Convergence Arise in the Brain?

Macroscale gradients have origins in the sensory processing system and travel up to higher-order cognitive regions like the TPJ. Information within these gradients is

hierarchical and becomes more abstract as it becomes distanced from sensory areas. As the gradients follow an axis of hierarchical abstraction and move away from primary cortices, a geographical space issue occurs: Gradients run into each other. This is where convergent brain areas seemingly occur, maximally distant from the primary sensory areas. Three processing constraints would need to be present for convergent processing streams to naturally lead to novel, higher-order functions processed in convergence zones and not in other parts of the brain: (1) “hierarchy” - information must be processed in hierarchies originating in primary sensory areas, (2) “abstraction” - distance from primary sensory areas should be correlated with the degree of abstraction, and (3) “equally and maximally equidistant” - sensory processing streams must terminate at polymodal points equally and maximally distant from primary sensory cortices.

Hierarchy: Information Is Processed in a Hierarchical Manner from Primary Sensory Areas to Higher-Order Cognitive Regions

The human brain follows a semimodular hierarchical organization (Bullmore & Sporns, 2009; Meunier, Lambiotte, Fornito, Ersche, & Bullmore, 2009; Zhou, Zemanová, Zamora, Hilgetag, & Kurths, 2006; Van Essen & Maunsell, 1983; Hubel & Wiesel, 1962), in which early neurons are simple and later neurons are more complex, clusters of neurons form functional modules, and these modules are hierarchically organized. To illustrate this process, we use a relatively well-established example of hierarchical organization in the brain, the visual cortex (Figure 1A). When we look at a human face, the primary visual cortex (V1) processes low-level information, such as edges of a particular orientation, in a particular location. These representations are then fed to higher visual areas that connect neighboring edges into lines and curves (V2), textures (V3/V4), and so on, until a holistic face is represented (lateral occipital face area). This is a computationally reasonable strategy that is in line with models of computer vision (Szeliski, 2011) and deep learning convolutional neural networks (Eickenberg, Gramfort, Varoquaux, & Thirion, 2017; Khaligh-Razavi & Kriegeskorte, 2014; Yamins et al., 2014). Hierarchical processing in the visual stream becomes less about processing the spatial characteristic, as calculation of motion becomes the dominant function in higher-order visual regions, that is, predicting an object’s motion energy from one moment to the next. In social processing, motion processing aids the detection of emotional expressions and eye movements (STS; Zhang, Japee, Stacy, Flessert, & Ungerleider, 2020; Engell & Haxby, 2007; Narumoto, Okada, Sadato, Fukui, & Yonekura, 2001). This type of predictive processing is extended into the TPJ, where the predictive inference is no longer about the energy of motion (e.g., representing the dynamic qualities of emotion), but how the mental state of another person might change in an abstract space.

This mental state prediction is accomplished by incorporating past experiences from memory, shifting attention from one internal representation to another, and making predictions of the next step (e.g., thinking about your friend's next move on a chessboard) as opposed to focusing on the sensory elements (e.g., processing the movement of your friend's hand, lifting a chess piece).

In summary, what started out as simple lines and facial forms have transformed into novel, higher-order, abstract information. Other sensory processing streams, such as auditory (Kikuchi, Horwitz, & Mishkin, 2010; Wessinger et al., 2001; Merzenich & Brugge, 1973) and somatomotor processing (Vaessen, Abassi, Mancini, Camurri, & de Gelder, 2019; Iwamura, 1998; Iwamura, Tanaka, Sakamoto, & Hikosaka, 1993), also follow this pattern of hierarchical abstraction (Figure 1B and C). These patterns are supported by neuronal recording in animals (Merzenich & Brugge, 1973) and neuroimaging in humans (Sepulcre et al., 2010). This principle of hierarchical organization is present in all of the converging processing streams, which leads to the question: How does information change along this hierarchy?

Abstraction: The Abstractness of a Representation Increases with Distance from Primary Sensory Areas

Within these information processing hierarchies, we alluded to increasing complexity of representation. We will make that relationship more explicit in this section. As representations move farther away from primary sensory cortices, they become domain-general, less detailed, and absent of physical features, resulting in abstracted higher-order representations, such as social intentions, memories of the past, or linguistic connotations. In fact, this notion was proposed early on by Mesulam (1998), who theorized that sensory inputs become abstracted as unimodal gradients converge toward the transmodal cortex. Building on Mesulam's theory, we hypothesize that the close spatial proximity of hierarchically abstracted representations is the key reason we observe higher-order cognition reliably within convergence zones.

The role of abstraction in these hierarchical gradients is to deliver information in an efficient manner. Returning to the example of visualizing a face, what started out as minute details of edges and orientations in V1 are later combined to represent more complex features (Croner & Kaplan, 1995; Van Essen & Maunsell, 1983). Eventually, this combination of information is enough to represent a human face. However, combining information in raw form is expensive, because of storage space and energy cost. It is important to note that the metabolic cost of information obtained is not a linear relationship: increasing the information rate (defined by firing rate of different axon caliber) twofold requires more than doubling the energy costs, demonstrating a law of diminishing returns (Balasubramanian & Sterling, 2009; Perge, Koch, Miller, Sterling, & Balasubramanian, 2009). Therefore,

abstraction provides a more compact representation of information contained at lower levels, increasing the density of information and providing a simple means to combine complex representations. In addition to abstraction from area to area, there are strong indications of increasing abstraction across the brain as a whole. For example, when semantic representations are mapped onto the cortical surface (Huth, de Heer, Griffiths, Theunissen, & Gallant, 2016), a gradient emerges, which begins with concrete words of perceptual/physical attributes (e.g., "fingers" and "four") and extends to abstract words of social and emotional attributes (e.g., "child" and "despised"). This concrete–abstract gradient aligns with the same neural hierarchy of unimodal sensorimotor regions and transmodal higher-order areas described above.

Adding another line of evidence to the abstraction argument, Fernandino et al. (2016) mapped words of sensory-motor attributes (color, shape, manipulation, sound, and motion) and found clear dimensions of increasing abstraction (Figure 2D). Each sensory attribute was associated with activation in its corresponding sensory-motor region, whereas the conjunction of these attributes, such as color/sound/motion, were mapped primarily onto zones of convergence. These findings support the concrete–abstract gradient along the sensory hierarchical axis.

This localization of abstract representations in convergence zones is nontrivial. Although it is generally accepted that abstraction increases in a posterior-to-anterior gradient (Taren, Venkatraman, & Huettel, 2011), the distribution of abstract information to several convergence zones (both anterior and posterior) found in a meta-analysis by Desai, Reilly, and van Dam (2018) was not expected. In fact, the convergence framework predicts that the distribution of abstract representation will occur in these areas maximally distant from primary sensory-motor regions, where macroscale gradients converge.

Maximal/Equidistance: Convergence Zones Are Maximally and Equidistant from Primary Sensory-motor Areas

Hierarchical abstraction cannot continue forever in limited brain space. We can therefore predict where convergence occurs. First, gradients converge at points equidistant from primary sensory cortices; any point that lies on the perpendicular bisector of, for example, the visual and auditory cortices, will be equidistant from these cortices, following the properties of a triangle. Second, gradients also converge at a point maximally distant from primary sensory-motor cortices. In fact, Margulies et al. (2016) found a spatial topology in which higher-order regions are not only maximally distant from sensory regions but also equally distant from each primary sensory cortex. The argument for processing gradients is supported by evidence of the paths seen as traveling waves of activity spread across the cortex (reviewed in Muller, Chavane, Reynolds, & Sejnowski, 2018; Roberts et al., 2019).

Furthermore, this topology of maximal distance and equidistance aligns with the concrete-to-abstract gradient. Margulies et al. (2016) show that aligning topic term reverse inference activation maps from neurosynth reveals precisely this type of hierarchical organization of abstraction (Figure 3). Again, using geodesic distance, primary sensory cortices and regions closer to the sensory origins were correlated with concrete terms (e.g., “motor” and “eye movement”), whereas regions maximally and equidistantly located from sensory origins, that is, higher-order regions, were correlated with abstract terms (e.g., “emotion” and “social cognition”). Furthermore, using both the equidistance maps from Margulies et al. (2016) and the gradient of abstract semantic representation from Huth et al. (2016), Huntenburg, Bazin, and Margulies (2018) demonstrate clear alignment between geodesic distance and concrete–abstract gradients. In other words, gradients that do converge will not only be farthest away from sensory cortices but will also have the most abstract, complex content. This further corroborates the idea that information in macroscale gradients becomes more abstracted with distance from sensory origins. Being located on a maximal-equidistant point from primary sensory cortices and being situated at the apex of the concrete-to-abstract gradient creates a rich setting for information to emerge once the gradients do converge.

The overlap between equidistance of gradients and hierarchical abstraction is unlikely to be a coincidence.

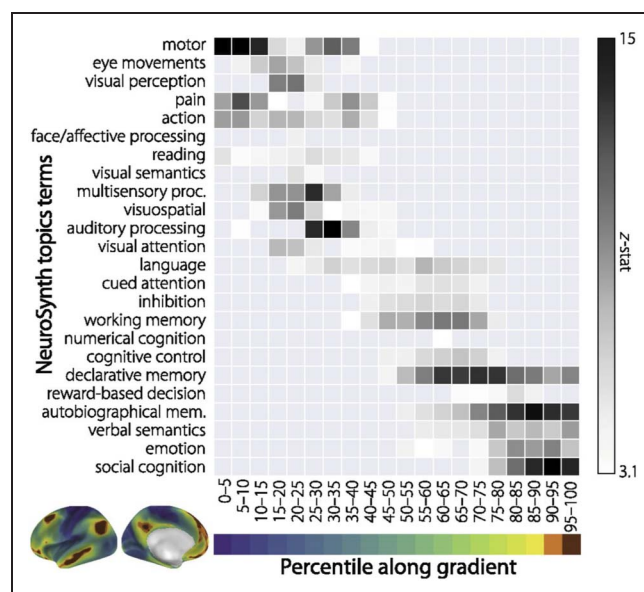


Figure 3. Axis of hierarchical abstraction of information aligns with the gradient of geodesic distance within the brain (figure adapted from Margulies et al., 2016). Percentile along gradient indicates geodesic distance from primary sensory origins to convergence zones. Neurosynth topic terms are sorted by the weighted mean of the location along the geodesic spatial gradient. Sensory terms, such as motor or eye movement, are associated with the lower end of the geodesic gradient (i.e., closer to sensory regions), whereas abstract terms, such as autobiographical memory or social cognition, are associated with the upper end of the geodesic gradient (i.e., convergence zones).

We have outlined evidence for a macroscale structure in the brain, consisting of processing streams of increasing abstraction and hierarchy. As the cortical area is limited, these streams will eventually converge. This presents an opportunity: Computationally expensive representations can be combined. Also, with this convergence, feature-rich, novel functions arise.

What Are Characteristics of Convergent Representations?

There is substantial evidence that feature-rich stimuli are processed at the intersection of converging processes. First, we review evidence of multisensory processing in the TPJ as a first indication of intersectional response. Second, we illustrate evidence of multisensory representations combined with higher-order cognitive processes in the TPJ. Lastly, we review evidence of higher-order representations that are combined with other domains of higher-order processes and propose that the TPJ plays a central role in constructing feature-rich representations, such as future state prediction.

Representations within Convergence Zones Are Multisensory

One example of intersectional representations in the TPJ is multimodal integration across sensory domains (Driver & Noesselt, 2008; Beauchamp, 2005; Beauchamp et al., 2004; Calvert, 2001). The TPJ is critical for integrating multisensory (Downar, Crawley, Mikulis, & Davis, 2000) and multispatial information, which involves a number of sensory domains, such as visuospatial attention, proprioception, visual gravitational motion perception, and heading perception (Kheradmand & Winnick, 2017). However, the integration of feature-rich information is not limited to the sensory domain.

Representations within Convergence Zones Are Integrative, with High-Order Cognition and Multisensory Facets

There is evidence of multisensory information being integrated into higher-order cognitive processes. In a study asking participants to encode both unimodal stimuli (either a visual drawing or auditory stimuli) and multimodal stimuli (a “train” presented as a whistle in conjunction with a film of a steam engine), the TPJ was significantly associated with the multimodal episodic retrieval process, whereas unimodal episodic retrieval only engaged sensory regions (Tibon, Fuhrmann, Levy, Simons, & Henson, 2019; Bonnici, Richter, Yazar, & Simons, 2016). In a recent review, the TPJ is identified as critical for integrating multimodal sensory details and mnemonic details from the past, which is essential in building atemporal contextual representations for episodic processing (Ramanan, Piguet, & Irish, 2018). These results already suggest that cognitive

processes in the TPJ are not simply single processes of attention or memory, as discussed in the sections “Getting Insight from the TPJ” and “Single Cognitive Domains Intersect the TPJ”. In fact, single processes are in stark contrast to combinatorial functions to which we now turn.

Representations within Convergence Zones Are Integrative Products of Higher-Order Processing Streams

Integration of single processes leads to combinatorial, higher-order cognition. One of the most common examples of this complex representation is mentalization, that is, projecting into another person’s mental state or intentions (Saxe & Kanwisher, 2003; Gallagher et al., 2002). Studies find that mentalization reliably activates convergence zones, including the mPFC, PCC, inferior frontal gyrus, and TPJ (Schurz et al., 2020). Mentalization forms the basis for numerous complex social behaviors, such as deception and strategic interaction, primarily in the TPJ. Future state prediction, another example of complex representations in the TPJ (Xu, Yuan, & Lei, 2016; Mars et al., 2012; Spreng, Mar, & Kim, 2009), shares parallels with mentalization, in that both projection beyond the current temporal state and interpolation toward an unforeseen state are necessary. This requires integration of past memories and cues from the current state, enabled by convergence. We next describe an example of each combinatorial function in convergence zones, enumerating the combination of single processes integral to each task.

Novel Functions that Arise in Convergence Zones

We have described convergence zones as producing representations that are combinations of the surrounding processing streams, but what sort of functions become possible when this higher-order information can be combined? A series of combinatorial functions have been

attributed to the TPJ, including deception, strategic mental state inference, and future prediction. We refer to these functions as “novel,” compared with our evolutionary ancestors and nonhuman primates, linked with the evolutionary expansion of convergence zones. Novel functions are hypothesized to have been made possible by the convergence of macroscale gradients within this evolutionarily expanded region. As opposed to single cognitive functions surrounding the TPJ, we see activation in the center of the TPJ—combinations of memory, attention, object recognition, and language—that may be the product of converging processing streams. Functions such as “autobiographical,” “response inhibition,” and “social” are likely to emerge in the center of the TPJ, which is evident in the reverse inference maps of neurosynth (Figure 4). In summary, gradients allow information to be passed to the next hierarchical stage and become integrated with surrounding information, providing a means for novel functions to arise.

Deception, one of the novel functions in human primates, requires an individual to predict the hidden states of an opponent, which is made possible through the combination of multiple single processes in the TPJ. A key study investigating deception utilized an economic bargaining game (Bhatt, Lohrenz, Camerer, & Montague, 2010), in which the “buyer” might deceive the “seller.” Only individuals who actively engaged in deception—strategically suggesting “deceiving” prices to throw off the seller’s gauge—were found to have correlated activation in the TPJ. Deception could be composed of a number of single functions: (1) paying attention to the current value (“attention/working memory”), (2) calculating the price likely to result in larger gains (“numeric calculation”), (3) anticipating the opponent’s mental reaction (“mentalization”), and (4) predicting the outcome if deception is successful. All of these are cognitive processes of what have traditionally been thought to be single functions of the TPJ; however, deception as a cognitive process is

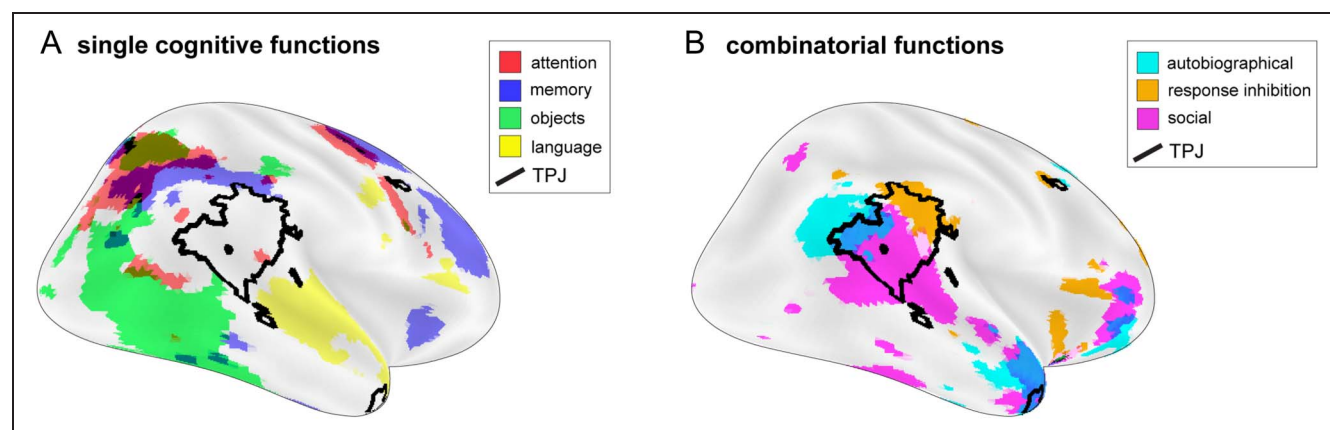


Figure 4. Functions in the TPJ are combinatorial. (A) Single cognitive functions, such as attention, memory, and objects, surround the TPJ. (B) In contrast, higher-order combinatorial functions, such as autobiographical, social, and response inhibition, are activated within the TPJ (Each map is a reverse inference map of the particular keyword, highlighted in the legend. Each map was downloaded from Neurosynth on December 2020, with a FDR-corrected p value of .01).

rather an integration of these single functions. Only through this integration do we see activation in the TPJ.

Similarly, the TPJ was activated in a simplified poker game, where participants were asked to bluff or fold in a series of consecutive games in the face of the same opponent (Carter et al., 2012). Bluffing in a poker round involves recalling your own and opponent's past behavior, strategic thinking of the card in your hands, and simulating the opponent's thoughts, that is, second-order beliefs. A cross-correlational analysis demonstrates that the opponent's moves from previous trials did influence the participant's decision on the current trial ("memory"). In conjunction, the TPJ was uniquely predictive of differentiating bluff trials not only against a human versus computer opponent ("social") but also for the more worthy opponent ("attention") (Figure 5A). In fact, no predictive value was found in the TPJ when either the memory, social, or attention components were missing. Moreover, when paired with other brain regions, the TPJ was the only region that reliably added classification performance of bluffing versus fold trials, indicating that the TPJ is the region uniquely contributing to complex deceptive behavior. Such combinatorial functions were also found in the attention-to-memory model (Cabeza et al., 2008, 2011; Ciaramelli et al., 2008), which illustrates the interaction between top-down/bottom-up attentional networks with memory processes, giving rise to novel functions in the TPJ (Figure 5B).

Interestingly, what happens during these series of gameplay and deception involves the participant not only

projecting themselves into the opponents' mental state but also temporally projecting into the future. Questions that may occur to the participant during the deception experiment or the strategic gameplay setting involve: "What might the opponent do if I bet?" "Will the opponent figure out that I was deceiving them?" "What will the opponent do in the next round, if they figure out that I was bluffing?" The common component demonstrated by these thought processes can be boiled down to future state prediction, an extremely feature-rich space. It is important to acknowledge that neurons are (broadly speaking) predictive. However, the TPJ incorporates an aspect of temporal prediction, that is, how the stimulus or state space is changing, which is different from the standard prediction of the presence or absence of a stimulus. In other words, in conjunction with memory, attentional, and social processes that are recruited during deception, the participant temporally projects themselves into the future and simulates what the opponent will think.

In fact, future state prediction, or prospection, is a function associated with the TPJ and other convergence zones. Accumulated research suggests that the posterior default mode network (DMN), including the TPJ, is implicated in future-oriented thought (Mars et al., 2012; Xu et al., 2016; Spreng et al., 2009). A causal link of TPJ and future state prediction was identified by Soutschek, Moisa, Ruff, and Tobler (2020) via continuous theta-burst stimulation, demonstrating that disrupting TPJ function disrupts the ability to think about future events as opposed to past events. In a mental time travel task,

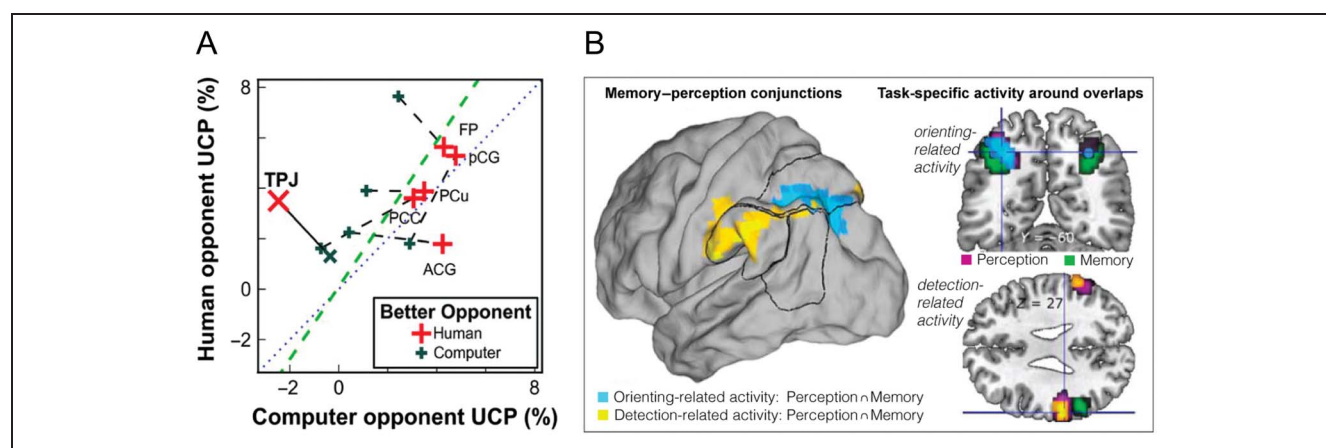


Figure 5. Unique functions emerge in the TPJ, a result of interactions of multiple cognitive processes. (A) The TPJ carries out a combination of functions (uniquely predictive of social opponents worthy of attention) rather than performing functions in isolation (either attention or social). In a study where participants were engaged in strategic gameplay, predictive information for each human and computer opponent was analyzed (Carter et al., 2012). Data points on the dotted-black midline indicate a brain region with equal predictive information for both human and computer opponents, notably, the TPJ is distant from the midline. This indicates that the predictive power is higher only when 1) the opponent is human, 2) the opponent is considered better, and 3) past trial histories are considered. (B) A combination of attentional and memory functions is found in the TPJ. Cabeza et al. (2011) conducted a 2×2 study of memory and perception tasks. In the memory task, participants would "orient" to the first word of memorized word pairs and "detect" the last word of the pair, whereas in the perception task, one would "orient" to the first consonant of letter streams and "detect" the vowel. The main effect of detecting/orienting, collapsed across perception and memory tasks, are plotted in the left panel. The right panel figure depicts activation of orienting ("right top") and detection-related activity ("right bottom") in each of the perception ("red") and memory ("green") tasks. In accordance with the convergence framework, each blob represents a unique combinatorial function consisting of what was previously thought of as separate processes.

participants were prompted to imagine either a past event (“undergraduate event”) or a future event (“Olympics in four years”). With theta burst stimulation to the TPJ, participants’ reaction time increased when responding to future versus past events, thus showing difficulty in simulating events on a temporal scale.

In summary, these feature-rich representations—such as deception, mentalization, and future state prediction—involve heavy computations. Combinatorial representations like these are likely constructed from incoming and neighboring information, again, through neuronal properties, such as wiring constraints (i.e., the probability of closer neurons being wired together) and traveling waves (i.e., neuronal activation that spreads to neighboring neurons). Via these neuronal properties, once gradients do converge, a mixture of single cognitive processes would lead to novel feature-rich representations. With the cascade of information integration, the combined information in the TPJ will be processed into a massive multifaceted space, resulting in higher-order functions that are novel.

The TPJ is not the only zone of convergence. A number of regions—such as the vmPFC, anterior temporal lobe, inferior frontal gyrus, dorsal mPFC, and PCC—share predicted properties of convergence zones. For instance, according to the “self-in-context” model, the vmPFC integrates interoception and perception into long-term narratives about the self, thereby creating conceptual mental models of the self in context (Koban, Gianaros, Kober, & Wager, 2021). A subset of these convergence zones is reliably identified as part of the DMN. The DMN is critical for functions, such as self-referential thought (Schacter, Addis, & Buckner, 2007), autobiographical memory, and temporal navigation (Addis, Wong, & Schacter, 2007), representations that are less related to immediate sensory information. This can, in part, be explained by the topographical characteristic of the DMN; they are located farthest away from primary sensory regions and are thereby less influenced by the external environment (Smallwood et al., 2021). For instance, researchers have found that DMN activation increases when task performance is dependent on rule-based learning or schema representations (as in a Wisconsin Card Sorting Task; Vatansever, Menon, & Stamatakis, 2017) as opposed to processing the immediate sensory input. For the DMN and TPJ alike, these convergence zones are capable of processing higher-order cognitive processes because of their characteristic topology, integrating hierarchically abstracted information from multiple incoming gradients.

Convergence Zones Show Evolutionary Expansion, Late Development, and Heterogeneity

As previously discussed, convergence—enabled via hierarchical abstraction and equidistance (Figure 6A)—allows novel combinatorial functions to emerge once macroscale gradients converge. The progressive abstraction of representations in convergence zones is reminiscent of changes

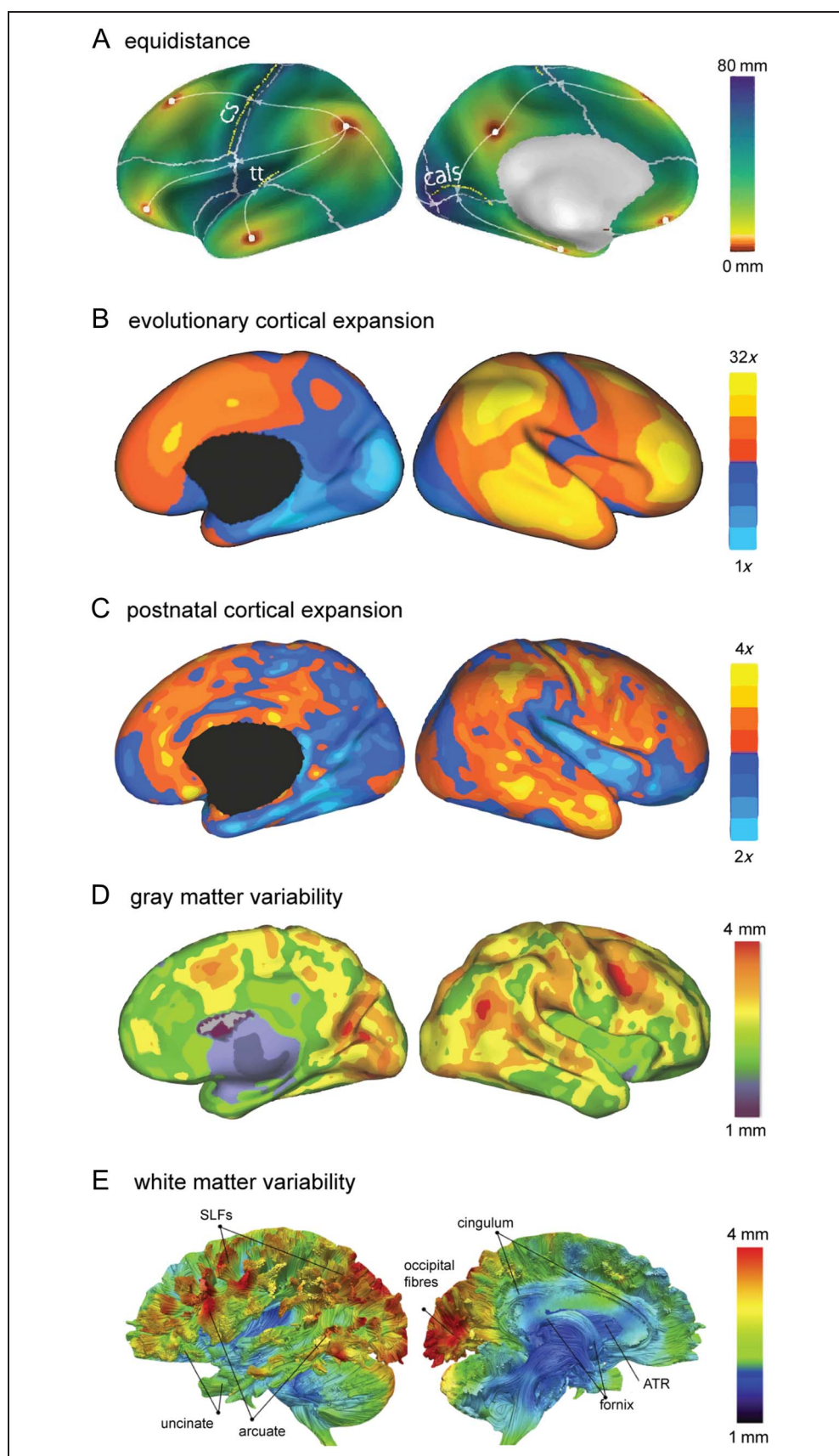
that occur over time in evolution and development and reflects heterogeneity across individuals.

Evolutionarily, convergence zones are expanded in humans compared with other species (Hill et al., 2010; Figure 6B). This expansion partially explains the enhanced role that cognitive functions play in adaptive human social behavior (Patel, Sestieri, & Corbetta, 2019). In fact, research beyond the TPJ has demonstrated that structural and functional properties of convergence zones correlate with the size of one’s social network, supporting the idea that convergence zones are critical for human social behavior (Noonan, Mars, Sallet, Dunbar, & Fellows, 2018).

Developmentally, the function of convergence zones emerges relatively late (Figure 6C). Using vision as an example, infants tend to respond to high-contrast edges before more complex shapes, distinguishing biological movement by 6 months (Woodward, 1998). Continuing this progression, successful mental state inference occurs much later (3–5 years), depending on the task used (Hughes & Leekam, 2004). As such, brain areas farther away from primary sensory cortices take longer to fully form. In fact, convergence zones, including the TPJ, undergo the largest cortical expansion after gestation (Fjell et al., 2015; Hill et al., 2010), continuing to develop until young adulthood. The late development of these regions mirrors their evolutionary expansion as well as their place in the computational hierarchy. This late development has additional side effects. Because they develop after a certain amount of world experience, they are less tied to genetic influence and more related to formative experiences.

Across individuals, therefore, convergence zones should be the most variable both anatomically and functionally. In fact, the TPJ is one of the most heterogeneous areas in the brain (Croxson, Forkel, Cerliani, & Thiebaut de Schotten, 2018; Mueller et al., 2013; Van Essen & Dierker, 2007; Figure 6D, E), in conjunction with other convergence zones such as the pFC. This idea is in line with the tethering hypothesis suggested by Buckner and Krienen (2013), which posits that associative cortices are regions with relatively fewer constraints in terms of development, supporting more variability. This variability may also play a significant role in the development of special interests in autism (Carter, Jung, Reaven, Blakeley-Smith, & Dichter, 2020), as convergent processing streams develop at different rates and result in different combinations of processing streams within convergence zones of individuals with autism. For example, relatively advanced development of processing in the dorsal attention stream could lead to a special interest in calculation, a common special interest in autism. In summary, the convergence account provides a framework for understanding the development of the TPJ and other convergence zones, which ties together disparate findings that span from equidistance from sensory cortices, late development, heterogeneity across individuals, and expansion in humans from evolutionary ancestors.

Figure 6. The evolutionary and developmental properties of convergence zones. Convergence zones—including the TPJ, ventral medial prefrontal cortex, anterior temporal lobe, inferior frontal gyrus, dorsal medial prefrontal cortex—are (A) equidistant from each other (Margulies et al., 2016), most expanded both (B) evolutionarily and (C) postnatally (Hill et al., 2010), and more variable in (D) gray matter and (E) white matter (Croxson et al., 2018). (Figures are presented in the original platform as published, therefore hemispheres may differ from subfigure to subfigure. Furthermore, we acknowledge functional/anatomical differences between hemispheres, however, consider it to be beyond the scope of this paper.)



Conclusion

The convergence framework provides a unifying view for predicting the location and function of higher-order regions. Information within the brain becomes abstracted and hierarchical along macroscale gradients that originate from primary sensory cortices. Convergence of such macroscale gradients brings feature-rich representations into close proximity, presenting an opportunity for novel functions to arise. The resulting zones of convergence exhibit a number of common features: equidistant from primary sensory cortices, developing late within an individual, and heterogeneous across individuals. The convergence framework provides a means to bring separate areas of cognitive psychology into alignment with computational and wiring constraints in accordance with evolutionary and developmental principles. For future directions on testing complex processes within the convergence framework, response distributions would be expected within convergence zones from tasks that require information from surrounding processing streams. Given knowledge of the contributing processing streams, these predictions could be used to design carefully targeted experiments that reveal novel, multifaceted functions in the brain.

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Author Contribution

Heejung Jung: Conceptualization; Visualization; Writing—Original draft; Writing—Review & editing. Tor D. Wager: Writing—Review & editing. R. McKell Carter: Conceptualization; Project administration; Supervision; Visualization; Writing—Original draft; Writing—Review & editing.

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Diversity in Citation Practices

Retrospective analysis of the citations in every article published in this journal from 2010 to 2021 reveals a persistent pattern of gender imbalance: Although the proportions of

authorship teams (categorized by estimated gender identification of first author/last author) publishing in the *Journal of Cognitive Neuroscience* (*JoCN*) during this period were M(an)/M = .407, W(oman)/M = .32, M/W = .115, and W/W = .159, the comparable proportions for the articles that these authorship teams cited were M/M = .549, W/M = .257, M/W = .109, and W/W = .085 (Postle and Fulvio, *JoCN*, 34:1, pp. 1–3). Consequently, *JoCN* encourages all authors to consider gender balance explicitly when selecting which articles to cite and gives them the opportunity to report their article's gender citation balance. The authors of this article report its proportions of citations by gender category to be as follows: M/M = .611; W/M = .179; M/W = .116; W/W = .095.

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